

Pitch Estimation Using Harmonic Product Spectrum derived from DCT

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Abstract—Estimation of pitch from a given segment of speech plays an eminent role in various speech processing applications, such as speech coding, speech recognition, speaker recognition tasks, speech synthesis, etc. Even though, there are several efficient algorithms, estimation of pitch frequency from speech signals that are severely degraded by noise is still a challenging task. In this paper, we propose a robust framework for pitch estimation using harmonic product spectrum (HPS) derived from discrete cosine transform (DCT) of the signal. This novel method exploits the better decorrelating nature of the DCT spectrum that enables the pitch harmonics to appear sharper in its spectrum. Potentially, this facilitates accurate pitch estimation at lower order of the harmonic product spectrum when compared with DFT-based HPS. Systematic evaluation is carried out to analyze the performance of the proposed method in comparison with some of the successful algorithms, like DFT-based HPS, SIFT, and Cepstrum-based technique. The results clearly show that the proposed algorithm outperforms the other algorithms for speech signals that are severely corrupted by noise (low SNR). The effectiveness of this method for different durations of analysis window, various orders of HPS, and the refinements are also discussed.

Index terms: Pitch, DCT, Harmonic product spectrum.

I. INTRODUCTION

Pitch (fundamental frequency) estimation plays a central role in various application areas such as speech recognition, speech synthesis, modeling of prosody and its modification, speaker recognition, and low-bit rate speech coding. In voiced speech, pitch is defined as the rate of vibration of the vocal folds. The vibrations of the vocal folds cause excitations to the vocal tract that are reflected as high energy regions in the speech signal. The time interval between two excitation points in the speech signal is defined as pitch period. Estimation of the pitch from the speech waveform is a challenging task due to quasi-periodic nature of the speech signal. The area of the pitch estimation has attracted a lot of researchers resulting in number of algorithms that includes both time-domain and frequency-domain techniques [1] [2]. However, in the case of noisy speech, the performance degradation is significantly high. This degrades their usefulness in many practical applications.

SIFT (Simple Inverse Filtering Tracking) algorithm is a well known time-domain technique for pitch estimation [3]. This method mainly depends on the LPC analysis and autocorrelation function (ACF). Several algorithms based on ACF

with various refinements are proposed in the literature [4] [5]. Though this method is proven to be very successful for clean speech, the major limitation is the sensitiveness of the ACF to the spectral envelope that causes pitch errors. Moreover, it is not very robust when the interference is strong as the autocorrelation function exhibits few spurious peaks that affect the accuracy of pitch estimation. The cepstrum-based pitch estimation is a commonly used frequency-domain technique. This involves the simple idea of picking the peak in the cepstral domain [6]. This method is seriously influenced by the size of the analysis-window over which the cepstrum is computed.

Among the various frequency-domain techniques, pitch estimation using the Harmonic product spectrum(HPS) derived from DFT spectrum of the signal is claimed to be relatively robust to noise [7]. The motivation for using HPS for pitch estimation is that the spectrum contains a series of peaks corresponding to the fundamental frequency and harmonic components at integer multiples of the fundamental frequency [8]. Here, the original spectrum is multiplied with compressed/down sampled spectra so that the peak at the fundamental frequency only gets magnified.

In our work, a novel method to characterize the source or to estimate the pitch is proposed, that computes HPS from the DCT of the signal. The motivation to use the DCT spectrum rather than DFT spectrum is that DCT decorrelates the signal well in the transform domain. Due to this, the peaks corresponding to the fundamental frequency and the higher harmonics are relatively sharper and greater in amplitude. Therefore, the HPS computed from the DCT spectrum is expected to give a stronger peak at the fundamental frequency location at lower order of the HPS itself. The number of down sampled spectra used in the computation of the HPS is defined as order in this work. This way of attaining the pitch information from DCT-HPS at the lower order will reduce the computational complexity. Further, this proposed method of pitch estimation is expected to be robust to noise and yield greater accuracy.

The organization of the paper is as follows. In Section II, the proposed method of pitch estimation using harmonic product spectrum derived from DCT spectrum in comparison with the DFT spectrum is presented. Section III details the experimental setup, refinements carried out to the proposed method and the performance analysis in comparison with some of the existing methods. Finally, in section IV, the important advantages of our technique are discussed.

II. PROPOSED METHOD

The HPS-based pitch detection is the simplest method that performs well under a wide range of SNRs. The HPS derived from DFT is used to measure the maximum coincidence of the harmonics in each spectral frame to estimate the fundamental frequency. We have proposed to use the DCT spectrum to compute the HPS to estimate pitch frequency for speech signals having low signal to noise ratio. The theoretical justification/motivation for this proposal is discussed below.

A. DCT spectrum vs. DFT spectrum

DCT is Discrete Cosine Transform that is same as symmetric DFT. As the DFT of an even symmetric function is always real, DCT has the same spectral envelope with that of DFT so that it exhibits all the formant and pitch structure that are present in the DFT spectrum. The DCT (specifically, we have used DCT form II) of a signal, say $x(n)$, is defined as

$$X_{DCT}(k) = \beta(k) \sum_{n=0}^{N-1} 2x(n) \cos\left(\frac{\pi k(2n+1)}{2N}\right) \quad (1)$$

where,

$$\begin{aligned} \beta(k) &= 1/\sqrt{N}; \quad k = 0 \\ &= \sqrt{2/N}; \quad 1 \leq k \leq N-1 \end{aligned}$$

In Equation (1), N is the length of $x(n)$ and k varies between 0 and $N-1$. It is well established that, like Karhunen-Loeve Transform, the DCT also decorrelates the signal well in the transform-domain when compared to other transforms [9]. This property causes the peaks corresponding to the fundamental frequency and the higher harmonics to be confined, in the sense that the peaks are sharper when compared to that of DFT-based magnitude spectrum. Due to this, the unwanted peaks in the transform-domain due to the presence of uncorrelated noise does not alter the locations of the peaks much, in the harmonic product spectrum. This clearly paves a way to accurate pitch estimation.

B. DCT-HPS vs DFT-HPS

Conventionally, the pitch estimation algorithm using the HPS derived from DFT spectrum measures the maximum coincidence of harmonics for each spectral frame as given in equation 2.

$$Y(e^{j\omega}) = \prod_{r=1}^R |X(e^{j\omega r})|^2 \quad (2)$$

$$p = \underset{\omega}{\operatorname{argmax}} \{Y(\omega)\} \quad (3)$$

The resulting periodic correlation array $Y(e^{j\omega})$, is searched for a maximum value as shown in equation 3 and the frequency, corresponding to the maximum is considered as pitch frequency. We are proposing to use the DCT spectrum to compute the HPS instead of DFT spectrum for estimating the pitch.

In the DCT spectrum, the peaks, corresponding to the fundamental frequency and the higher harmonics, are sharper

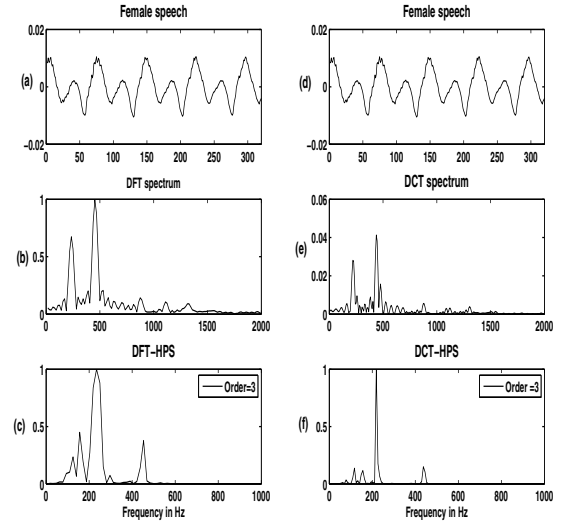


Fig. 1. Comparison of HPSs computed with order 3 (a),(d) Female speech signal with 216Hz pitch frequency (b) DFT spectrum (c) DFT-HPS (e) DCT spectrum and (f) DCT-HPS.

with higher magnitude. The down sampled/compressed spectra of DCT allow the higher harmonics also to be located at the fundamental frequency location. Thus, multiplying the original spectrum with the compressed spectra magnifies the peak at the fundamental frequency. Though the same thing can be achieved using the DFT spectrum while computing HPS, the number of compressed spectra (called as order), required to yield a sharper peak at the fundamental frequency location alone is comparatively lesser than that of the DFT-HPS. This facilitates the pitch estimation at reduced computational cost. Fig. 1 shows the HPSs derived from DFT and DCT for a female speech with order as 3. The actual pitch frequency that is manually estimated for the female speech considered (fig. 1(a) and 1(d)) is 216Hz. The DFT-HPS computed with order 3 shown in fig. 1(c) has a broader peak only at 234Hz. Further, other peaks present in the HPS are also significantly larger. However, the DCT-HPS computed at the same order shown in fig. 1(f) has a narrower peak exactly at the fundamental frequency location and also the peaks at other frequency locations have lesser magnitude. These figures clearly show the scope of accurate pitch estimation at lower order and in turn, at reduced computational cost with the DCT-HPS when compared to DFT-HPS. This is expected to be true in most of the voiced speech segments and systematic evaluation is performed to confirm this.

In order to analyze the performance of the DCT-HPS approach, initially we have considered the voiced segments of 5 male and 5 female speech which are free from noise. Both the DCT-HPS and DFT-HPS are computed over these segments with a fixed order. The frequency corresponding to the maximum peak in the HPSs are declared as the estimated pitch frequencies. The estimated pitch is matched with the manually estimated reference pitch and marked as correct if the error in estimation is less than 20% of the manually estimated pitch frequency. The experiments are repeated with varying orders in the range 1 to 8 and are plotted in the fig. 2. The

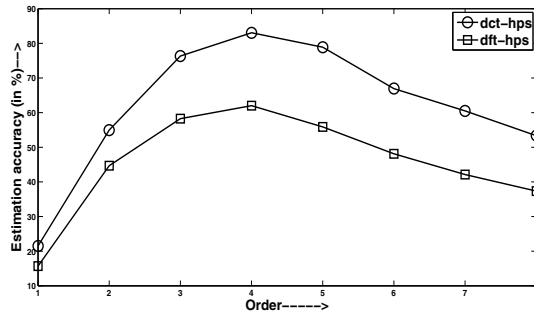


Fig. 2. Performance comparison of DFT-HPS and DCT-HPS over clean speech with respect to order.

estimation accuracy achieved by DCT-HPS is observed to be consistently better than that of DFT-HPS for a wide range of orders. Though this novel method shows favorable results over clean speech when compared to DFT-HPS, the major claim of the proposed method is that it will be robust to noise. Since the peaks in the DCT spectrum are so confined, the influence of the noise at the narrow peaks in the spectrum are expected to be minimal. This will improve the accuracy of pitch estimation even for speech signals that are severely corrupted by noise which is explored in this work.

III. PERFORMANCE ANALYSIS AND RESULTS

A. Experimental set-up

TIMIT corpus is used to evaluate the proposed method of pitch estimation and to compare the results with some of the existing techniques. The analysis involves 5 male and 5 female speech data from the TIMIT corpus. Some of the well known pitch estimation techniques namely, SIFT algorithm, Cepstrum-based pitch estimation method, and the technique using DFT-HPS are implemented to carry out comparative performance analysis. In this work, speech signal is segmented into frames of 20ms each. The effect of noise on the accuracy of pitch estimation methods is evaluated by adding noise to the speech data. Gaussian white noise is generated and added with the speech signal at different strengths/levels in order to generate speech data of varying SNRs in the range -10db to +10db. The estimation accuracy of the proposed method in comparison with all the other techniques under consideration are analyzed using the speech data with varying SNRs. The reference pitch information for evaluation is obtained by calculating the pitch from the manually marked instants of excitation of the signal. The time interval between two adjacent instants of excitation is actually the pitch period which is defined in section I. The performance of all these algorithms was evaluated only for the voiced segments using the reference pitch. The estimated pitch for a frame is decided to be correct if the error between the estimated and actual value is less than 20%.

B. Performance Enhancement

The speech signals corrupted by noise are considered for evaluating the performance of the proposed method under noisy condition. The algorithm for estimating the pitch using DCT-HPS is given below:

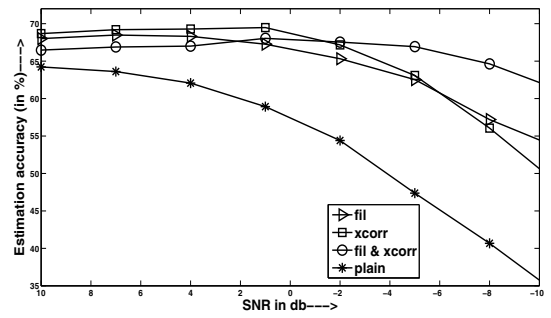


Fig. 3. Best method to compute DCT-HPS for both male and female speech

- Preprocess the noisy speech data suitably (as explained below).
- Compute N -point DCT (here, $N = 1024$) over the speech segments to obtain the DCT spectrum.
- Derive the HPS from the DCT spectrum with the order set as 5.
- Declare the frequency corresponding to the maximum value in the HPS as the pitch frequency.

Various preprocessing algorithms are tried on the speech signals prior to the computation of DCT to handle the noise. The preprocessing done to suppress the noise shows significant improvement in the performance of the proposed method. The two major preprocessing techniques applied are low-pass filtering and autocorrelation. Noise signal primarily comprises higher frequency components that can be easily eliminated by low-pass filtering the noise-corrupted signal. However, the low frequency noise still prevails and impacts the performance of pitch estimation for which autocorrelation is suggested. The advantage of taking the autocorrelation of the speech signal is that, it emphasizes the periodicity in the signal even if it is corrupted by random interference [10]. However, computing ACF alone is not favourable for pitch detection when the interference is strong for the reasons discussed in section I.

In this work, the autocorrelation function (ACF) is applied in the preprocessing stage and the causal portion of it alone is taken for further processing as ACF is an even function. Experiments are carried out to analyze the proposed method with various combinations of these two preprocessing techniques as mentioned below: (i) Linear filtering followed by DCT (ii) Autocorrelation followed by DCT (iii) Linear filtering followed by autocorrelation followed by DCT (iv) Direct computation with no preprocessing stage.

Analysis is carried out with noise-corrupted speech signals having varying SNRs and the accuracy of estimation is recorded. Fig. 3 shows these comparative performances. Computing the DCT-HPS directly without any preprocessing and using it for pitch estimation gives 65% accuracy at high SNR (10db) and gets drastically reduced to 35% when SNR is decreased. The DCT-HPS computed after filtering alone or only autocorrelation performs significantly better than direct computation. However, their individual performances are very close for the whole range of SNR. The DCT-HPS computed

over filtered and auto-correlated signal is observed to give results that are marginally better than the others at high SNRs and substantially better at low SNRs. Hence, the performance of the proposed method can be enhanced by choosing the appropriate pre-processing technique as linear filtering followed by autocorrelation prior to the computation of DCT-HPS.

C. Comparative analysis

The SIFT algorithm, the cepstrum method, and the technique using the DFT-HPS are implemented to perform comparative analysis. The preprocessing in these methods is also preferred the same way as in the proposed method to perform a fair comparison. Fig. 4(a) and fig. 4(b) show estimation accuracy in percentage by the proposed algorithm in comparison with the existing methods for male and female speech. The proposed method of pitch estimation is proven to be extremely robust to noise, relative to all other methods. Cepstrum-based estimation is observed to be poorly performing when compared to all other algorithms. SIFT algorithm performs well for clean speech and slowly deteriorates when SNR reduces. The SIFT algorithm dominates the proposed method for speech signals up to 0db SNR (approx). However, the performance of the DCT-HPS accelerates greatly compared to SIFT for the signals that are severely corrupted by noise i.e. when SNR is reduced below 0db. Comparing DCT-HPS and DFT-HPS : DCT-HPS is always better than DFT-HPS for all male data. For female data, DFT-HPS slightly outperforms DCT-HPS for signals having high SNR and the DCT-HPS overtakes it when the signal quality is degraded by noise. To take the complementary information from the DFT-HPS, the product of DCT-HPS and DFT HPS is computed and is observed to yield slightly better results than DCT-HPS.

Further, the performance of this technique is analyzed with varying window sizes and orders. The speech signals having low SNR values in the range of -10db are considered for these analysis. The proposed algorithm gives better results when the order is set as 5 for both female files and male files. The performance goes down when the order is increased further which is illustrated in fig. 4(c). This is because the higher harmonics are eliminated by the low-pass filtering operation performed during preprocessing stage. Consequently, the compressed spectra do not have multiple peaks at the fundamental frequency position to strengthen it. This greatly affects the estimation performance. All the pitch estimation algorithms are tested for varying window sizes. It is observed that all the techniques are less sensitive to change in window size except for Cepstrum-based method (Fig. 4(d)).

IV. CONCLUSION

In this paper, a novel pitch estimation technique using harmonic product spectrum derived from the DCT of the signal is proposed. Using the better decorrelating nature of DCT, it is shown that the harmonic product spectrum derived from DCT spectrum have great potential in estimating the pitch than the DFT spectrum. Further, it has been shown that this technique is highly robust to noise compared to other existing algorithms. The proposed method clearly outperforms the SIFT algorithm, the well known method, for speech signals having low SNR. Apart from these, the computational complexity involved is

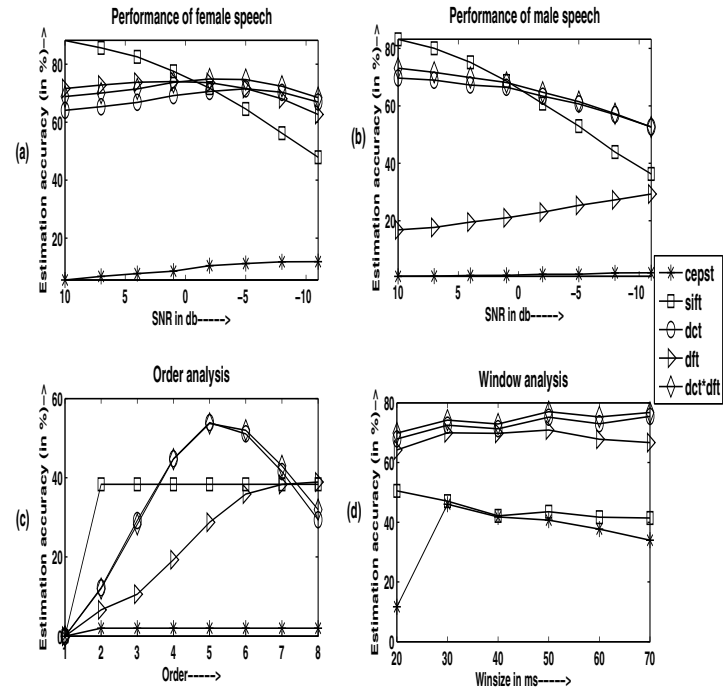


Fig. 4. Performance analysis (a) Estimation accuracy for female speech (b) Estimation accuracy for male speech (c) Order analysis (d) Window analysis

considerably less as accurate pitch estimation is made possible at lower order of the DCT-HPS compared to DFT-HPS-based technique.

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